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Smart Cloud-Based Medication Dispensing System for Clinical Use

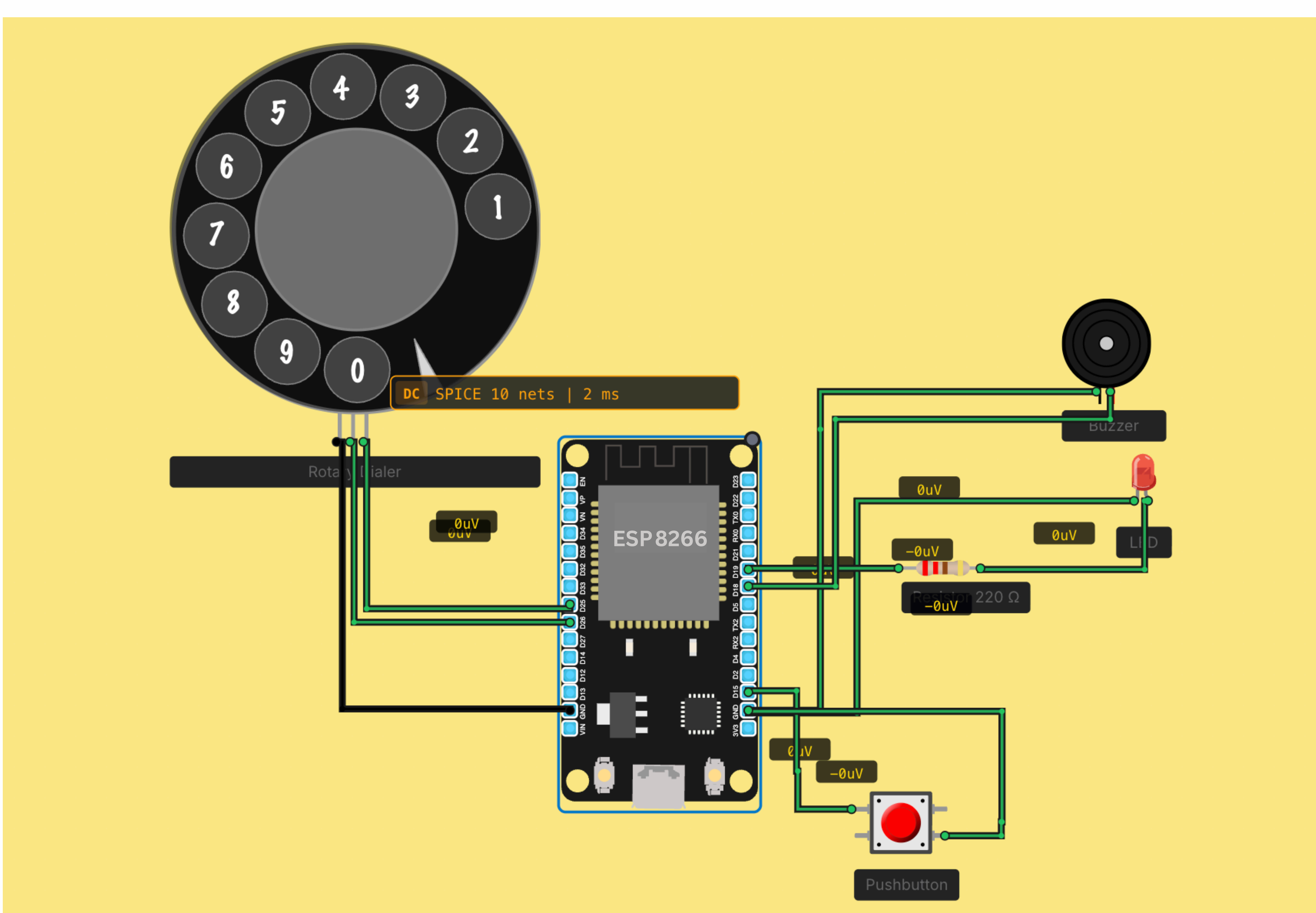
Abstract

This project presents a Smart Centralized Medication Dispenser for clinical environments where doctors manage multiple patients. The system combines. an ESP-based dispenser with a servo-driven rotating mechanism for precise pill release. A cloud platform built using Firebase enables secure role-based access (Doctor, Caregiver, Patient). Clinicians can remotely schedule and monitor medication, reducing human error and improving efficiency.

Methodology

Medication errors and missed doses remain major healthcare challenges, often caused by heavy workload and reliance on manual processes. Existing automated systems are expensive and lack the flexibility needed for diverse clinical environments.
This project introduces a low-cost IoT-based automated dispensing system using an ESP microcontroller and servo-controlled compartments, integrated with a cloud application via Firebase. By leveraging IoT connectivity, the system allows doctors, caregivers, and patients to manage medication schedules remotely in real time, providing a scalable and efficient solution for modern clinical settings.

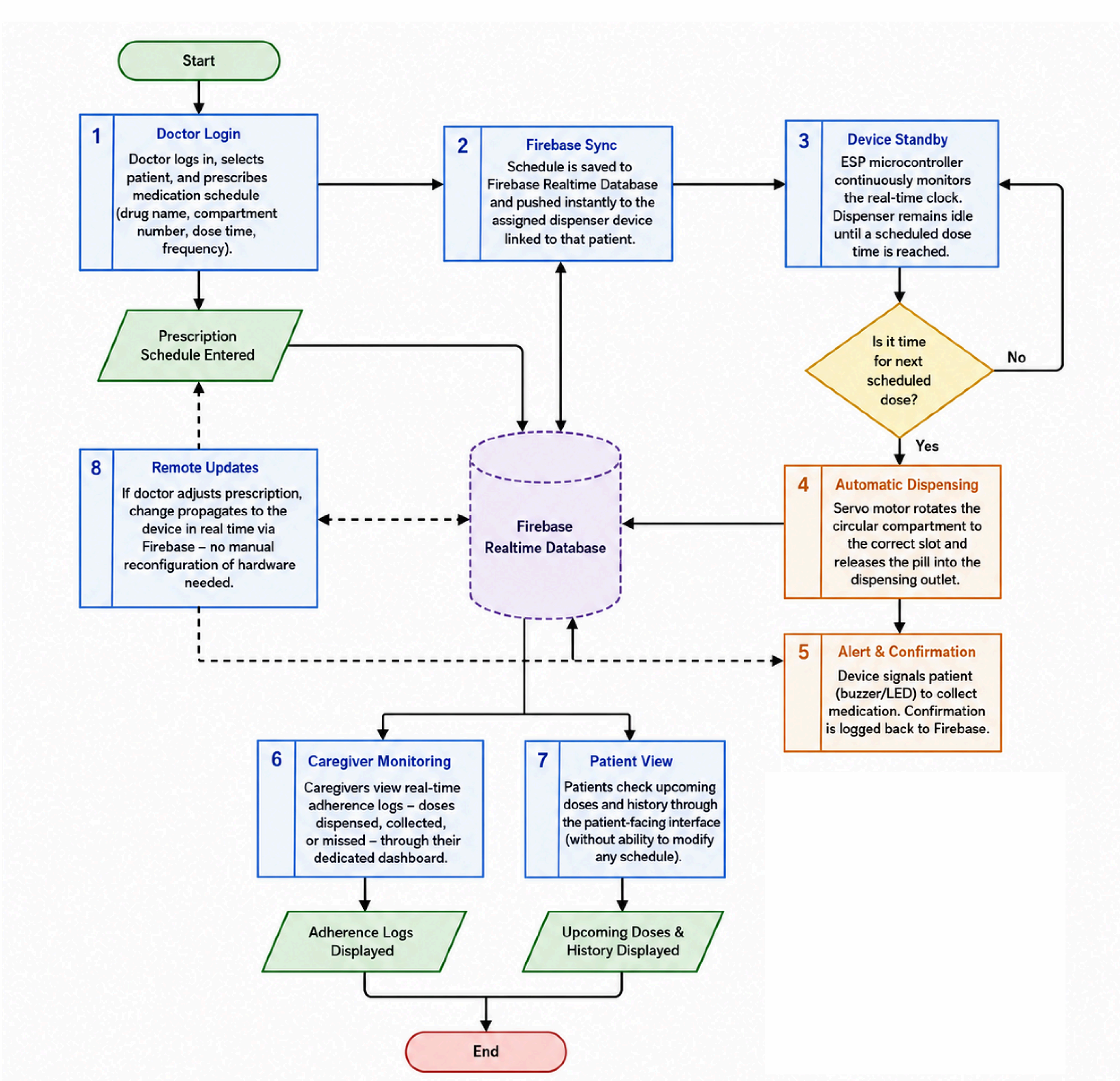
Circuit



Key Features

- Remote Control: Manage medication anytime via cloud interface
- Role-Based Access: Doctor → Caregiver → Patient hierarchy
- Low Cost: Affordable alternative to commercial systems
- Real-Time Synchronization: Instant updates between hardware and software

Work Flow



Conclusion

The proposed ESP-based system successfully replicates key functions of high-cost clinical dispensers at a significantly lower price. The structured role hierarchy ensures secure control of medication schedules, while cloud synchronization via Firebase maintains consistency between hardware and software.
Future improvements include multi-device control, refill monitoring, missed-dose notifications, and audit logs.
This system demonstrates the feasibility of scalable, affordable smart healthcare automation.